THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 12: **Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The total surface area of a solid hemisphere of radius r is: 1

(a) $2\pi r^2$ **(b)** $3\pi r^2$ **(c)** $4\pi r^2$ **(d)** πr^2

2. The volume of a solid made of a cone (radius r, height h) on a hemisphere of radius r is: 1

(a) $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$ **(b)** $\frac{1}{3}\pi r^2 h + \frac{4}{3}\pi r^3$ **(c)** $\pi r^2 h + \frac{2}{3}\pi r^3$ **(d)** $\frac{1}{3}\pi r^2 (h + r)$

3. Two cubes each of edge 5 cm are joined end to end. The surface area of the resulting cuboid is: 1

(a) 250 cm² **(b)** 300 cm² **(c)** 150 cm² **(d)** 200 cm²

4. A cylinder and a cone have equal radii and equal heights. The ratio of their volumes is: 1

(a) 3 : 1 **(b)** 1 : 3 **(c)** 2 : 1 **(d)** 1 : 1

5. A student states: "When two solids are joined, the total surface area of the new solid is the sum of their total surface areas." The correct response is: 1

(a) Correct
(b) Incorrect — the surfaces in contact are not counted
(c) Correct only for cubes
(d) Cannot be determined

6. The volume of a hemisphere of radius 21 cm is: 1

(a) 19404 cm³ **(b)** 38808 cm³ **(c)** 9702 cm³ **(d)** 2772 cm³

7. A toy is a cone (radius 3 cm, slant height 5 cm) on a hemisphere of radius 3 cm. Student A says its surface area is $\pi r l + 2\pi r^2 = 33\pi$; Student B says $\pi r l + 3\pi r^2 = 42\pi$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. When a solid sphere of radius r is cut into two hemispheres, the increase in total surface area is: 1

(a) πr^2 **(b)** $2\pi r^2$ **(c)** $3\pi r^2$ **(d)** $4\pi r^2$

9. The volume of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 1540 cm³ **(b)** 770 cm³ **(c)** 154 cm³ **(d)** 3080 cm³

10. The slant height of a cone of height 24 cm and radius 7 cm is: 1

(a) 25 cm **(b)** 24 cm **(c)** 31 cm **(d)** 17 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A cylinder of radius 7 cm and height 10 cm has a hemisphere of the same radius on top. Find the total volume of the solid. 2

12. A tent is a cylinder of height 3 m and radius 7 m surmounted by a cone of slant height 10 m. Find the area of canvas used (no floor). 2

13. A student claims that the surface area of the cuboid formed by joining two cubes of edge 4 cm is 192 cm². Justify or refute. 2

14. Find the volume of a toy in the form of a cone of radius 3.5 cm and height 6 cm mounted on a hemisphere of the same radius. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A juice glass is cylindrical with inner diameter 5 cm and height 10 cm, but its bottom has a raised hemispherical portion of the same diameter. Find the apparent and actual capacity of the glass. (Use $\pi = 3.14$) 3

16. A wooden pen stand is a cuboid 15 cm × 10 cm × 3.5 cm with four conical depressions, each of radius 0.5 cm and depth 1.4 cm. Find the volume of wood in the stand. 3

17. A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid. 3

OR

A hemispherical depression is cut out from one face of a cubical block of edge l such that the diameter of the hemisphere equals l. Find the surface area of the remaining solid. 3

18. An ice-cream cone is a cone of radius 3.5 cm and height 12 cm, filled and topped with a hemisphere of ice-cream of the same radius. Find the volume of ice-cream. 3

OR

A solid consists of a cylinder of radius 3 cm and height 8 cm with a cone of height 4 cm (same radius) on top. Find its total surface area. (Use $\pi = 3.14$) 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A bird-bath is in the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and its radius is 30 cm. 5
(i) Find the total surface area of the bird-bath (curved surface + bowl).
(ii) How many litres of water can the bowl hold?

OR

A solid is composed of a cylinder with hemispherical ends. The whole length of the solid is 108 cm and the diameter of the ends is 36 cm. Find the cost of polishing its surface at 7 paise per cm². 5

20. **Case study:** An ice-cream shop serves a cup that is a cylinder of radius 3 cm and height 5 cm, topped with a hemisphere of ice-cream of the same radius. (Use $\pi = 3.14$) 5
(i) Find the volume of the cylindrical part.
(ii) Find the volume of the hemispherical top.
(iii) Find the total ice-cream in one cup.
(iv) How many litres are needed for 50 cups?

OR

Case study: A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius. 5
(i) Find the slant height.
(ii) Find the surface area of the toy.
(iii) Find the cost of painting it at ₹0.50 per cm².
(iv) Find the volume of the toy.

END OF QUESTION PAPER — **Total: 40 Marks**